Myanmar Institute of Information Technology



Tour Guide Assistant for Bagan Pagodas

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Abstract

Bagan, Myanmar, is renowned for its thousands of temples and pagodas spread across a large area, making navigation challenging for tourists. This project develops a web-based Tour Guide Assistant that helps users efficiently plan routes between pagodas and landmarks.

The system features an interactive frontend built with HTML, CSS, and JavaScript, providing a user-friendly interface for selecting starting points, destinations, and viewing routes. The backend, developed in Python, handles routing logic, pathfinding computations using the A (A-Star) algorithm*, and communicates with the database through a RESTful API. A MongoDB database stores information about landmarks, route connections, and path statuses, enabling scalable and flexible data management.

By combining real-time pathfinding with detailed landmark information—including Ananda Temple, Dhammayangyi Temple, Alotawpyae Pagoda, and Shwezigon Pagoda—the system helps tourists save time, avoid confusion, and enjoy a smoother travel experience. The framework is scalable and can be adapted to other tourist destinations in the future.

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Chapter 1

Introduction

1.1 Background

Bagan, Myanmar, is an ancient city with thousands of pagodas spread across a vast area, making it challenging for tourists to navigate efficiently. Traditional guides and printed maps often fail to offer real-time, optimized navigation. With advancements in web-based technologies and AI algorithms, a smart route-planning system can significantly improve tourist experiences in such heritage zones.

1.2 Objectives

- To design and implement a smart algorithm for shortest pathfinding on a grid-based map.
- To help tourists easily navigate between landmarks in Bagan.
- To build a user-friendly web interface with real-time routing.
- To store and manage location and path data effectively using a database.
- To allow future scalability for other regions beyond Bagan.
- To help reduce tourist confusion, improve route efficiency, and enhance user travel experience in cultural heritage zones.

1.3 Scope

The system will be accessible through any modern web browser and optimized for both desktop and mobile devices. It is designed for tourists, tour guides, and cultural heritage enthusiasts visiting Bagan. Users will be able to view an interactive map, select pagodas to visit, and get the shortest route calculated using the A* algorithm. The platform will also provide historical and cultural information for each pagoda, enhancing both navigation and the learning experience.

1.4 Project Description

The Tour Guide Assistant for Bagan Pagodas is a web-based application designed to help users navigate Bagan's historical landmarks. It provides an interactive platform where users can select starting and destination points to find the shortest route between pagodas.

The application uses the A* (A-Star) algorithm to calculate the most efficient path, accounting for real-world distances and travel restrictions. The shortest route, along with details about each pagoda's history and visiting tips, is displayed on a dynamic map. Users can also customize tours based on time or walking distance, making the app a flexible guide for exploring the region.

The frontend is built with HTML, CSS, and JavaScript, while the backend, developed in Python, handles API requests and executes pathfinding calculations. The system stores data on landmarks, paths, and blocked routes in a MongoDB database, with a RESTful API connecting the components.

Ultimately, the Tour Guide Assistant for Bagan Pagodas enhances the tourist experience by automating route planning, reducing travel time, and providing an intuitive interface for simple navigation.

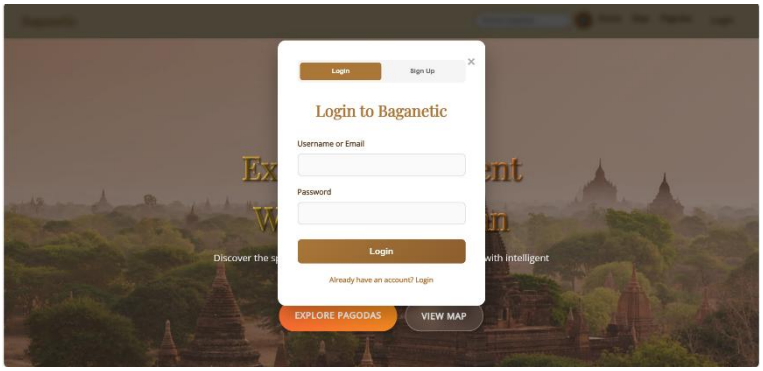


Figure 1.1 Login / Sign Up Page



Figure 1.2 Home Page



Figure 1.3 Pagodas Page



Figure 1.4 Map Page

1.5 Project Schedule

Table 1.1 Project Schedule

Tasks	July	August	September
Research and Software Requirements			
Project Outline			
Developing			
Midterm Report			
Midterm Presentation			
Adding Features			
Final Report and Presentation			

Chapter 2

System Overview and Components

2.1 System Overview

The Tour Guide Assistant for Bagan Pagodas integrates navigation, cultural information, and personalized route planning to enhance the tourism experience in Bagan. It combines mapping technology with intelligent pathfinding to provide an efficient and informative tour. Key features include:

- **User Registration and Authentication:** Secure login system to allow users to save preferences, customize tours, and access additional features.
- **Interactive Map and Pagoda Selection:** Users can browse and select one or more pagodas to visit from an interactive map of Bagan.
- **Optimal Route Calculation:** The A* algorithm is used to determine the most efficient path between selected locations, considering real-world distances and travel constraints.
- **Pagoda Information Access:** Detailed historical, cultural, and visiting tips are available for each pagoda.
- **Customizable Tours:** Users can adjust itineraries based on available time, walking distance, or areas of interest (architecture, history, spirituality).

2.2 Components

- **Frontend:** HTML, CSS, and JavaScript-based web interface with Leaflet.js for interactive map rendering, providing a responsive design for both desktop and mobile.
- **Backend:** Python-based server using a web framework (Flask or FastAPI, to be finalized) for handling requests, running the A* algorithm, and managing user sessions.
- **Database:** MongoDB for storing landmark details, route data, and user profiles.
- **Algorithm:** A* pathfinding algorithm for computing the shortest and most optimal travel paths between selected destinations.

The A* algorithm finds the shortest path by combining:

- **g(n)**: The cost from the start node to the current node.
- **h(n)**: The heuristic estimate of the cost from the current node to the goal.

The total cost is**: f(n) = g(n) + h(n)**

Example Calculation — Node: Ananda Pagoda

Start Node (S): Shwezigon Pagoda

- Coordinates: 21.18333° N, 94.89472° E

Goal Node (G): Alodawpyi (Alodawpyae) Pagoda

- Coordinates: 21.17667° N, 94.88250° E

Current Node (n): Ananda Pagoda

- Coordinates: 21.17058° N, 94.86662° E

g(n): Actual Cost from Start to Current Node

The distance from Shwezigon Pagoda to Ananda Pagoda is calculated using the Haversine formula:

g(n)=3.241 km

h(n): Heuristic Cost from Current Node to Goal

The straight-line (great-circle) distance from Ananda Pagoda to Alodawpyi Pagoda is:

h(n)=1.781 km

This heuristic is admissible because it never overestimates the actual travel distance.

f(n): Estimated Total Cost

f(n)=g(n)+h(n)

f(n)=3.241+1.781=5.022 km

Start: Shwezigon Pagoda — 21.18333° N, 94.89472° E

Goal: Alodawpyi Pagoda — 21.17667° N, 94.88250° E

Table 2.1 Distance Dataset using A* Algorithm

Node	Latitude (°)	Longitude (°)	g(n) km (Start→Node)	h(n) km (Node→Goal)	f(n) km=g+h
Shwezigon Pagoda	21.18333	94.89472	0.00	1.287	1.287
Alodawpyi Pagoda	21.16265	94.86629	1.287	0.00	1.287
Ananda Temple	21.16670	94.87281	3.241	1.781	5.022
Dhammayangyi Temple	21.16815	94.86025	2.693	1.592	4.285
Thatbyinnyu Temple	21.16548	94.86393	3.404	1.872	5.276
Htilominlo Temple	21.16705	94.86661	1.74	1.574	3.314
Sulamani Temple	21.16308	94.86857	2.154	1.343	3.497
Shwesandaw Pagoda	21.16369	94.86823	3.31	2.42	5.730
Gawdawpalin Temple	21.15348	94.85622	3.633	2.236	5.869

Chapter 3

System Design and Flowchart

3.1 System Design

The Tour Guide Assistant for Bagan Pagodas employs a modular architecture, allowing each component to be developed, maintained, and updated independently. The design includes:

component to be developed, maintained, and updated independently. The design includes:

- **User Interface Module:** Provides an intuitive, responsive web interface for users to view the interactive map, select pagodas, and customize their tours.
- **Route Calculation Module:** Implements the A* algorithm to determine the most efficient travel route based on user-selected destinations and travel constraints.
- **Database Layer:** Uses MongoDB to store and retrieve landmark details, route data, and user preferences, ensuring data integrity and fast queries.

3.2 Flowchart and picture of the system design

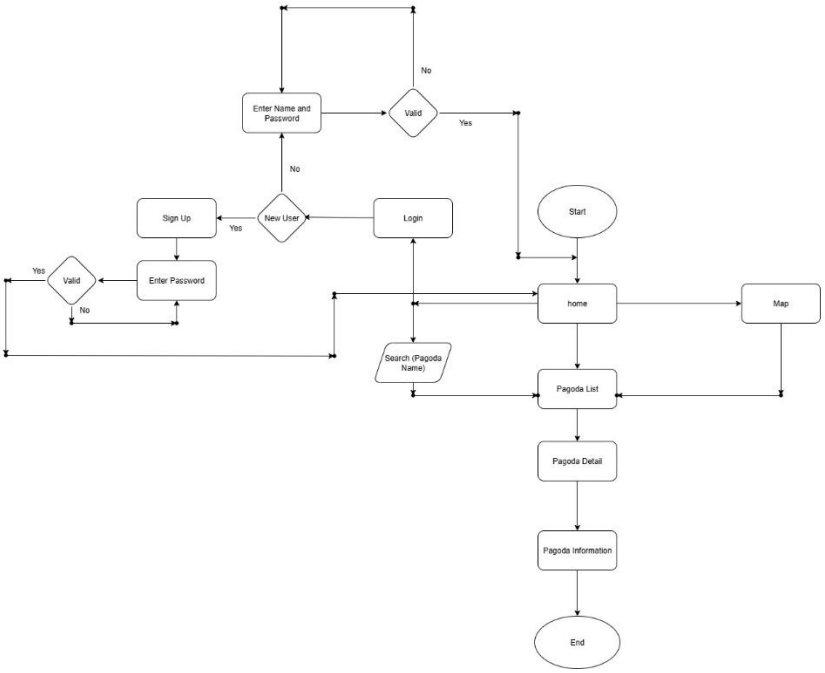


Figure 3.1. Flowchart of Project Description

Chapter 4

System Implementation

4.1 I**mplementation Steps:**

- **Setup** – Install Python, MongoDB and required libraries.
- **Data Prep** – Collect pagoda and road network data, clean, and store as nodes and edges in MongoDB with distance weights.
- **Backend** – Build APIs to fetch pagoda data, run shortest-path A* algorithm, and return route details.
- **Frontend** – Create a responsive web UI with Leaflet.js to select pagodas, display routes, and show distance.
- **Database** – Define collections (pagodas, nodes, edges) with 2dsphere index for geospatial queries.
- **Testing** – Perform unit, integration, and performance tests; fix bugs and optimize queries.

Chapter 5

Conclusion

5.1 Summary

The **Tour Guide Assistant for Bagan Pagodas** addresses key challenges faced by tourists and tour guides in navigating the Bagan Archaeological Zone. By providing an interactive map, real-time pathfinding using the A* algorithm, and detailed information about pagodas, the system simplifies route planning and enhances the overall tourism experience. The use of a web-based platform with a Python backend and MongoDB database ensures flexibility, scalability, and reliability. Users can efficiently plan multi-stop itineraries, avoid blocked or restricted paths, and explore Bagan's historic landmarks with confidence and ease.

5.2 Future Enhancements

- ⋆ **Mobile App Development:** Create native Android and iOS applications to allow offline navigation and greater accessibility.
- ⋆ **AI-Powered Recommendations:** Implement algorithms to suggest optimal pagoda tours based on user preferences, time, and distance.
- ⋆ **Multi-Language Support:** Include support for multiple languages to cater to international tourists.
- ⋆ **Integration with External Services:** Connect with APIs for real-time updates on weather, road conditions, or tourism events.
- ⋆ **User Feedback & Analytics:** Collect user feedback and analyze usage patterns to improve route suggestions and enhance the user experience.

References

★ **Leaflet JS : https://leafletjs.com/reference.html**

★ **MongoDB :** https://www.mongodb.com/docs/manual/